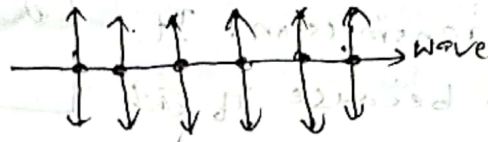


Polarization

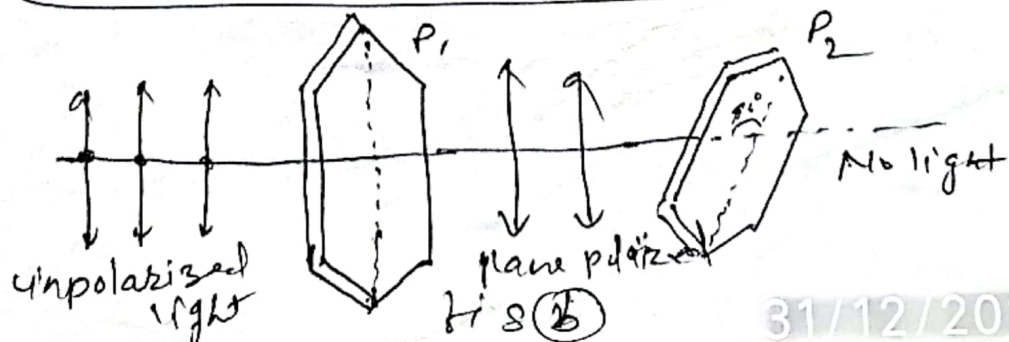
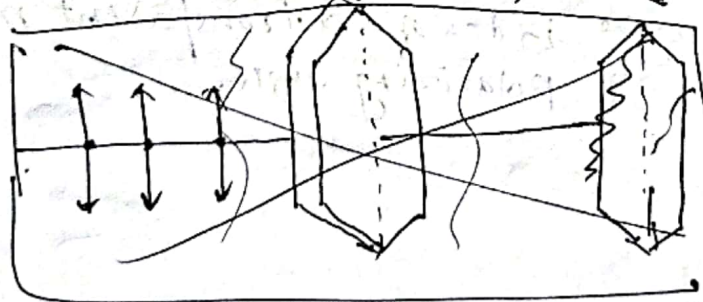
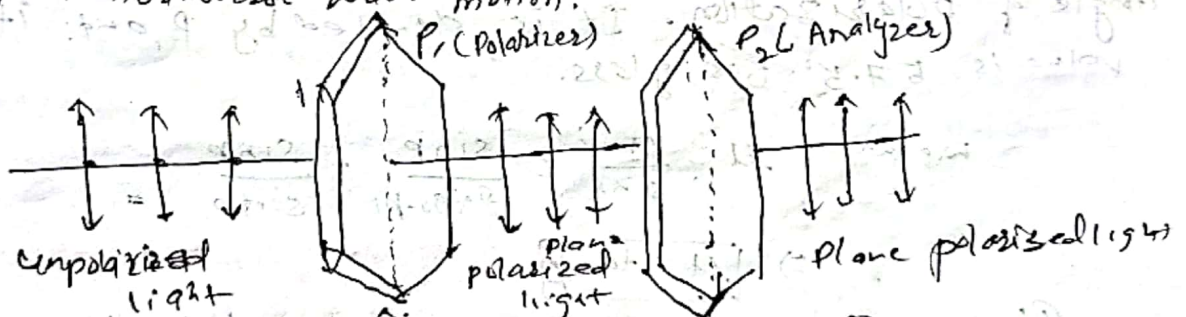
In natural light, millions of transverse vibrations occur in all the directions perpendicular to the direction of propagation of wave. But for convenience, we can assume the rectangular components of the vibrations with one component lying on the plane of the diagram and other perpendicular to the plane of the diagram.



• perpendicular to plane
↓ parallel to the plane

The phenomenon of confining the vibration of light wave to a single plane perpendicular to its direction of propagation is called polarization of light.

When ordinary light is passed through a pair of tourmaline crystal plates P_1 and P_2 with their planes at right angles to the directions of propagation of light as shown in fig (a), the intensity is maximum in this position. But, when the plane of P_2 is rotated through 90° as shown in fig (b), the intensity is minimum. This shows that light is transverse wave motion.



Polarization by Reflection

When a ray of light incident on the certain crystals, a part of light undergoes reflection and other part undergoes transmission. It is happened so because if each vibration of the light can be resolved into two components which are

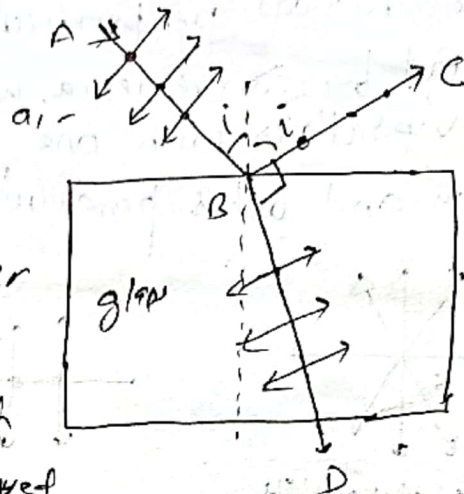


Fig: polarization by reflection

right angle to each other. The vibration which is parallel to the plane of crystal undergoes transmission whereas the vibration which is perpendicular to the plane of the crystal undergoes reflection. For the certain value of angle of incident both the reflected and transmitted part of the light undergo plane polarized, then corresponding angle of incident is called angle of polarization. It is denoted by p , and its value is 57.5° for glass.

$$\text{we know } \mu = \frac{\sin i}{\sin r} = \frac{\sin p}{\sin(90-p)} = \frac{\sin p}{\cos p}$$

$$\Rightarrow \boxed{\mu = \tan p}$$

Thus the refractive index of a transparent medium is equal to tangent of polarizing angle.

Double refraction: When the light through certain crystals like calcite, tourmaline, ruby etc. undergoes splitting into two components. Out of these two components: one obeys the law of refraction but other doesn't obey the law of refraction which are respectively called ordinary ray (o-ray) and extra-ordinary ray (e-ray). This phenomenon of light is called double refraction or birefringence.

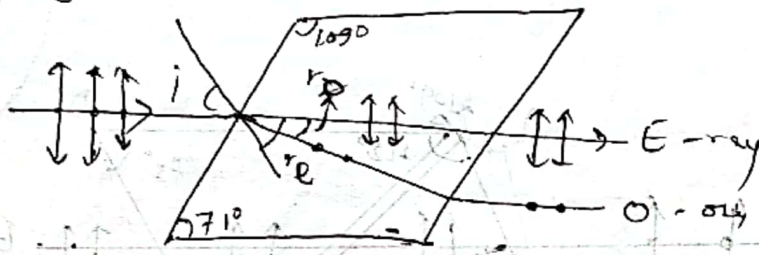


Fig: Phenomenon of double refraction in calcite crystal.

Refractive Index of ordinary ray, $\mu_o = \frac{\sin i}{\sin r_o} = \frac{c}{v_o}$

extra-ordinary ray & $\mu_e = \frac{\sin i}{\sin r_e} = \frac{c}{v_e}$

~~Since $v_e < v_o$, $\mu_e > \mu_o$~~

~~Since $v_o > v_e$~~

Now, these crystals in which $v_e > v_o$, then $\mu_o > \mu_e$. These types of crystals are called negative uniaxial crystal.

For example calcite, tourmaline, ruby etc.

Again those crystal for which $v_e < v_o$ (i.e. $\mu_o < \mu_e$), these crystal are known as positive uniaxial crystal eg. quartz, iron oxide etc.

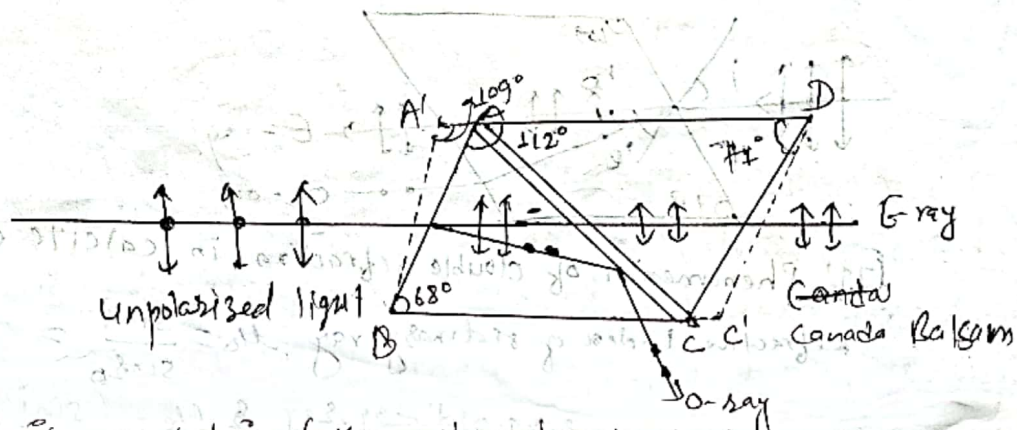
Nicol Prism: (William Nicol in 1829).

An optical device which is used for producing and analysing the plane polarized light is called Nicol prism.

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Nicol prism : It is an optical device made from calcite or quartz crystal and is used for producing and analyzing plane polarized light. When a ray of light is passed through calcite crystal it splits off into ordinary ray and extraordinary ray. Nicol prism is constructed in such a way that it eliminates O-ray by total internal reflection and allows E-ray to pass through it.

construction :



A calcite crystal is taken whose length is three times that of its breadth. The faces of crystal are cut down to reduce the angle at B and D from 71° to 68° . Now the crystal is cut down into two pieces and a transparent material called Canada Balsam is kept in two surfaces and joined.

working : Consider a ray of light incident on the Nicol prism. It will undergo double refraction into O-ray and E-ray. The E-ray is transmitted through the prism but ordinary ray suffer total internal reflection at the Canada balsam layer. It is because Canada balsam ($\mu_b = 1.550$) is optically denser than calcite for E-ray ($\mu_E = 1.486$) and rarer than calcite for O-ray ($\mu_O = 1.658$). The E-ray moving from rarer medium to denser medium is refracted at the Canada balsam layer and is transmitted through the prism. The O-ray moving from denser medium to rarer medium is totally internally reflected at the Canada balsam layer for the angle of incidence greater than the critical angle.

The value of critical angle for calcite to Canada balsam layer for O-ray is $\mu = \frac{\mu_b}{\mu_O} = \frac{1.550}{1.658} \Rightarrow C = 69.2^\circ$ Hence plane polarized light can be obtained from Nicol prism.

5 Huygen's Explanation of Double Refraction:-

According to wave theory of light every points of the primary wavefront are the sources for the secondary wavefront. On the basis of this assumptions Huygen explained the phenomenon of double refraction. According to him, when a secondary wavefront passes through certain crystal, which are capable to produce double refraction, the secondary wavefront splits into two components which are right angle to each other i.e. splits into o-ray and e-ray.

The velocity of o-ray is same in all directions and hence the wavefront of the o-ray is spherical in shape but that of e-ray is different in the different directions, and hence the wavefront of e-ray is ellipsoid.



spherical wavefront for o-ray



ellipsoidal wavefront for e-ray

Consider a source of light inside the crystal. When we consider negative uniaxial crystal, then the velocity of e-ray is found to be greater than that of o-ray; hence the wavefront of e-ray lies outside the wavefront of o-ray as in fig. (a). Likewise in the case of positive uniaxial crystal velocity of e-ray is less than that of o-ray; hence wavefront of e-ray lies inside the that of o-ray as shown in fig. (b).

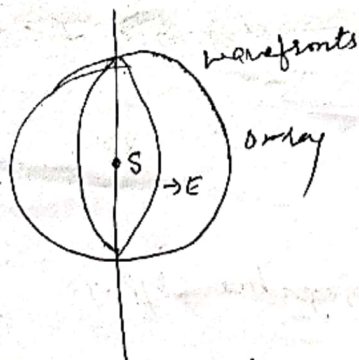


fig (a) positive uniaxial crystal

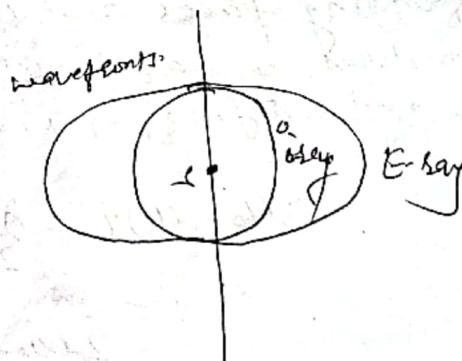


fig (b) negative uniaxial crystal

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6 Retardation plates (Retarders)

A crystal plate which is capable of producing double refraction is called ~~refrac~~ retardation plate. If it produces certain path difference between o-ray and e-ray while passing the thickness of the crystal, this plate is called wave plate.

Mainly retarders are: ① Quarter-wave plate

② Half-wave plate

1) Quarter-wave plate:

A quarter wave plate is a thin plate of double refraction crystal having the optic axis parallel to its refracting faces and its thickness adjusted such that it introduces a quarter-wave ($\lambda/4$) path difference (or phase diff of 90°) between the e-ray and o-ray propagating through it.

For negative crystal, path diff = $(\mu_o - \mu_e)t$

and, for positive crystal path diff = $(\mu_e - \mu_o)t$

$\therefore$ for quarter wave plate $p.d = \lambda/4$

$$\therefore (\mu_o - \mu_e)t = \frac{\lambda}{4}$$

$$\Rightarrow t = \frac{\lambda}{4(\mu_o - \mu_e)}, \text{ for negative crystal}$$

$$\text{and } t = \frac{\lambda}{4(\mu_e - \mu_o)}, \text{ for positive crystal.}$$

A quarter wave plate is used for producing elliptically or circularly polarized light.

2) Half-wave plate: A half wave plate is a thin plate of double refraction crystal having the optic axis parallel to its refracting faces and its thickness chosen such that it introduces a half-wave ($\lambda/2$) path difference between o-ray and e-ray. In this case.

$$(\mu_o - \mu_e)t = \frac{\lambda}{2}$$

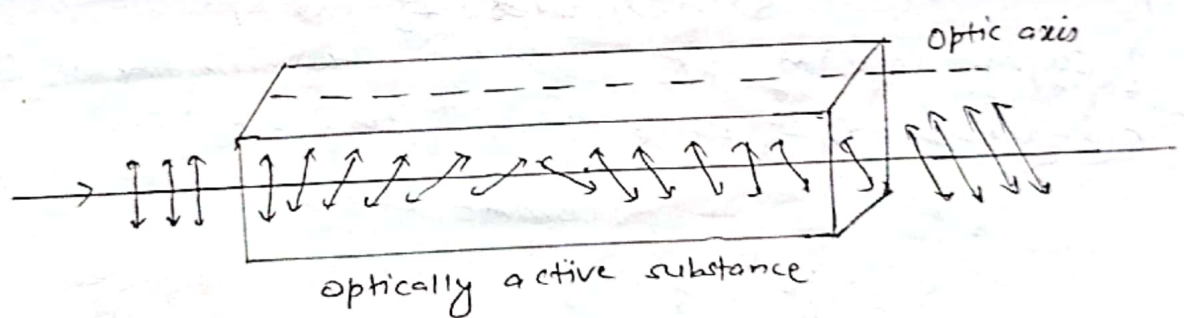
$$\Rightarrow t = \frac{\lambda}{2(\mu_o - \mu_e)} \text{ for negative crystal}$$

A half wave plate rotates the plane of polarization of the incident plane polarized light through an angle 2θ .

7
Optical activity: The phenomenon of rotating of plane of vibration by certain crystals or substance is known as optical activity and the substance is known as an optically active substance. It has been found that calcite does not produce any change in the plane of vibration of plane polarized light, so it is not optically active. Quartz and cinnabar are examples of optically active crystals while aqueous solutions of sugar, tartaric acid etc are optically active solutions.

Optically active substances are:

- i) Dextrorotatory substances; which rotate the plane of polarization of light towards right (clockwise) are known as right-handed or dextrorotatory.
- ii) Laevorotatory substance; which rotate the plane of polarization of light towards left (anti-clockwise) are known as left-handed or laevorotatory.



Specific Rotation:

A measure of the optical activity of a sample is the rotation produced for a 1 mm slab for a solid, or 100 mm path length for a liquid. This measure is called the specific rotation.

If an optically active material is kept between two crossed polarizers, the field of view becomes bright. In order to get darkness once again, the analyzer has to be rotated through an angle i.e. the angle of plane of polarization rotation. This angle depends upon:

- (a) thickness of the substance
- (b) density of material or concentration of solution
- (c) wavelength of light, λ
- (d) the temperature

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The amount of rotation θ caused by crystalline material is given by $\theta = \alpha l$

where α = rotational constant

In solution amount of rotation θ is,

$$\theta = Scl$$

where c = concentration

S = specific rotation.

The specific rotation for a given wavelength of light at a given temperature is defined as the rotation produced by one decimeter long column of the solution containing 1 gm of optically active material per cc. of solution.

$$[S]^t = \frac{\theta}{l \times c} = \frac{\text{rotation in degrees}}{\text{length in decimeter} \times \text{conc. in gm/cc}} = \frac{100}{l(\text{cm})C}$$

Numerical:

A 200 mm long tube and containing 48 cm³ of sugar solution produces an optical rotation of 11° when placed on a saccharimeter. If the specific rotation of sugar solution is 66°, calculate the quantity of sugar contained in the tube in the form of a solution.

$$\theta = 11^\circ, L = 200 \text{ mm} = 20 \text{ cm}$$

$$S = 66^\circ$$

$$S = \frac{10 \times \theta}{L \times c}$$

$$\therefore c = \frac{10 \times \theta}{S \times L} = \frac{10 \times 11}{66 \times 20} = 0.083391 \text{ cm}^3$$

$$V = 48 \text{ cm}^3$$

$$\therefore M = CV = 4 \text{ gm} \underline{\text{Ans}}$$